Skills, Distortions, and the Labor Market Outcomes of Immigrants Across Space

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Introduction

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  - Underperfoming in the labor market reduces social cohesion and influences aggregate outcomes
- **Existing evidence**: US immigrants and natives mainly **differ in**:
  i. **Job choices** → human capital (Lagakos et al., 2018), task specialization (Peri and Sparber, 2009), labor market barriers (Birinci et al., 2024)
  ii. **Residential choices** → preferences for locations (Albert and Monras, 2022)

  → Does not account for **interaction** among these choices
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- Documents **3 stylised facts:**
  i. The **earnings gap** b/w immigrants and natives is **larger in big cities**
  ii. No city-size earnings premia only for immigrants from **low-income countries**
  iii. Workers from **high-income countries** work more in **cognitive jobs**, especially in big cities

- Interprets these facts with a **spatial GE model** including:
  - Workers' heterogeneity: **human capital** and city-occupation **amenities**
  - Worker-specific local labor market distortions: **wedges** on the marginal product of labor
  - Cross-city heterogeneity: technology and housing supply

- Quantifies the **role of heterogeneity and wedges** on the earnings gap b/w
  - Immigrants and natives: human capital **-18.9%**, amenities **-6.2%** , wedges **-9.3%**
  - Big and small cities: human capital **+1.1%**, amenities **+3%** , wedges **-0.1%**

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Contribution to the Literature

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  New fact: spatial distribution of occupational choices differ by origins

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  iii. Workers from high-income countries work more in cognitive jobs, especially in big cities. Doubling the city size:
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From the Data to the Model

- **Data shows:**
  - Earnings gap increases with city size for immigrants from low-income countries
  - Workers from high-income countries work in cognitive occupations in big cities
- **A spatial equilibrium model to:**
  - Quantify how different factors affect job choices in U.S. cities b/w immigrants and natives
  - Study the consequences of inflows of new immigrants on earnings inequality
- **The model has three building blocks:**
  - Differences in technology across cities (Atalay et al. (2023), Eeckhout et al. (2024), Giannone (2023))
  - Workers' heterogeneity in skills and tastes for where to work and live (Schoellman (2012), Lagakos et al. (2018), Albert & Monras (2022))
  - Labor market distortions (Hsieh et al. (2019), Birinci et al. (2024))

The Model in a Nutshell

Environment

- **Static economy:** cities (local labor markets) and agents (workers)

- **Cities, production and housing:**
  - $j \in \{1, ..., J\}$ cities
  - Firm in city j produces Y_j with CES technology using human capital in two occupations $o \in \{M, D\}$
  - City-specific productivity bias θ_j in cognitive occupations D
  - Absentee landlords own land T_j and produce housing H_j

- **Workers:**
  - Continuum of workers $i \in [0, 1]$
  - Each worker i is endowed with human capital $\mathbf{s} = (s_M, s_D)$ and belongs to a group $g = (k, e, x)$
  - Each group has a measure ϕ_g s.t. $\sum_g \phi_g = 1$
  - Cobb-Douglas utility function in consumption and housing goods

$$U_{jog} = c_{jog}^{(1-\alpha)} h_{jog}^{\alpha} \mathbf{z_{jog}} \exp\{\varepsilon_{jo}\}$$

$\varepsilon_{jo} \sim$ Gumbel$(0, 1)$ i.i.d. taste shock, city-occupation amenities z_{jog}, α expenditure share in housing

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The Model in a Nutshell: Choice Equation

- **Firms**
  - Set skills prices r_{jo} to max profits and min costs Firm problem
- **A worker $i \in g$**
  - Earns: $w_{jog} = r_{jo} s_{og} \tau_{jog}$
    - τ_{jog} is a group-specific local labor market compensation wedge
  - Given their city-occupation choice, max utility subject to her budget constraint (earnings) Worker problem Indirect utility
  - The share of workers from group g choosing a city j and an occupation o is:

$$\pi_{jog} = \frac{\gamma p_j^{-\alpha} \overbrace{r_{jo} s_{og} \tau_{jog}}^{w_{jog}} z_{jog}}{\sum_{j' \in \mathcal{J}} \sum_{o' \in \mathcal{O}} \gamma p_{j'}^{-\alpha} \underbrace{r_{j'o'} s_{o'g} \tau_{j'o'g}}_{w_{j'o'g}} z_{j'o'g}}$$

- **Absentee landlords**
  - Housing supply is governed by: $p_j = \left(\frac{H_j}{T_j}\right)^{\frac{1}{\zeta_j}}$, H_j is the housing demand, T_j is land, ζ_j is housing supply elasticity Housing supply

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Model Identification and Calibration

From the Model to the Data: Assumptions and Identification

- **Identifying assumptions:**
  - Native workers are not subject to labor market distortions
  - $z_{jog} = 1, \forall g$ in the smallest city and non-cognitive occupation
- **Other assumptions:**
  - ζ_j, T_j do not vary across city
  - ϕ_g is given
  - $\tau_{jog} = \tau_{jok}$, i.e. wedges vary only by origin, location, and occupation
- **Dimensionality reduction:**
  - 2 cities $\rightarrow$ $\{\text{Small City}, \text{Big City}\}$
  - 3 countries of origin $\rightarrow$ $\{\text{Natives}, \text{Low-Income}, \text{High-Income}\}$
  - 2 education groups $\rightarrow$ $\{\text{No College}, \text{College}\}$
  - 3 experience groups $\rightarrow$ $\{0-14, 15-29, 30+\}$
- **Parameters:**
  - 6 externally calibrated Externally calibrated parameters
  - 100 calibrated using the MSM Identification and internal calibration

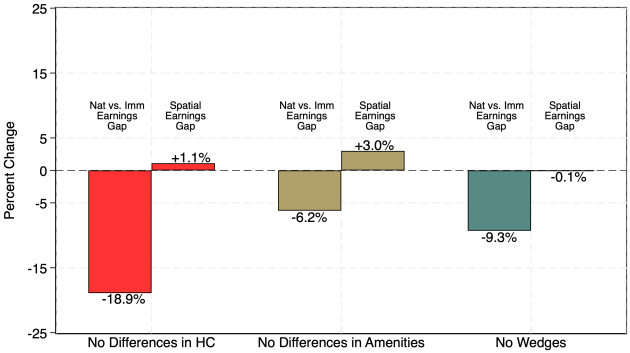
Counterfactual Exercises

The Model as a Laboratory

- **Quantification:** study the role of human capital, amenities and wedges on:
  - Earnings gap b/w natives and immigrants $\overline{w}^{\text{Gap}}_{\text{Workers}}$
  - Earnings gap b/w big and small cities $\overline{w}^{\text{Gap}}_{\text{Cities}}$ Gaps definitions

- **Counterfacutals:** for all immigrants, keeping fixed the other parameters, remove:
  1. Differences in **human capital** with natives
  2. Differences in **amenities** with natives
  3. **Wedges** on earnings
  4. Differences in amenities and wedges
  5. Differences in human capital, amenities and wedges

The Earnings Gaps: Human Capital vs Amenities vs Wedges



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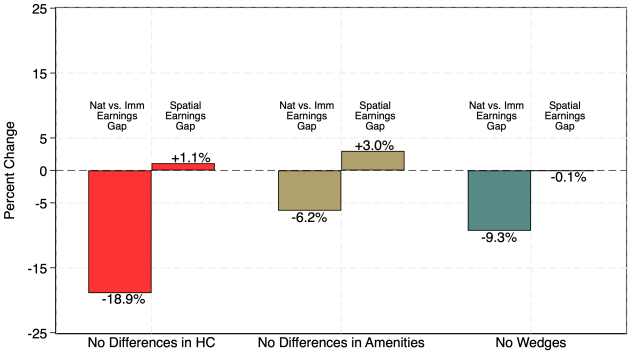

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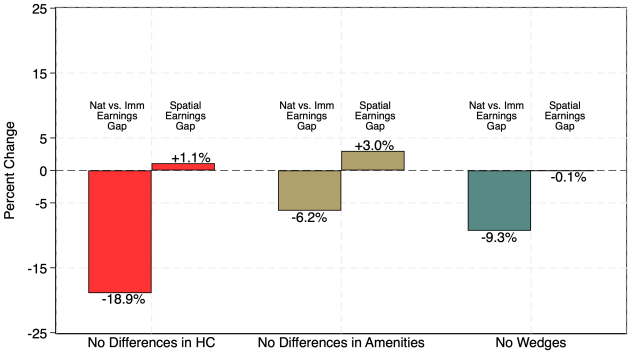

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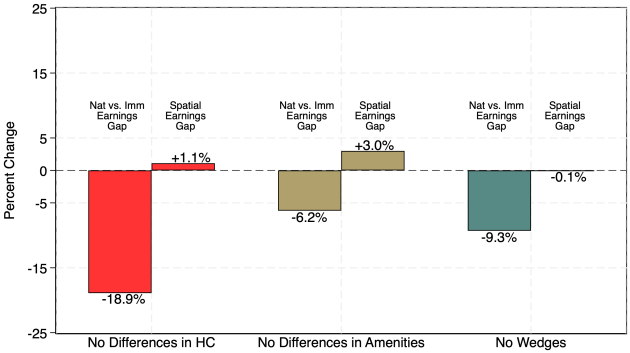

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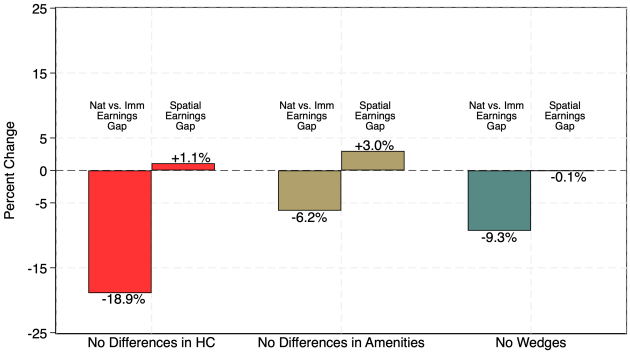

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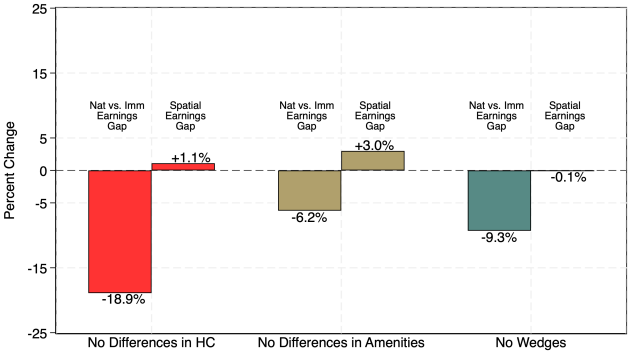

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Policy Exercises

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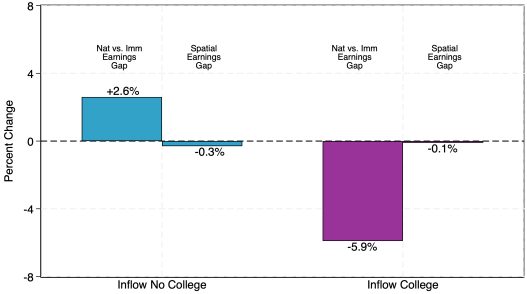
  - **Policy 2:** inflow of immigrants with college education



- Inflow of immigrants with **no college education**:
  - Earnings gap b/w immigrants and natives **+2.6%** vs spatial earnings inequality **-0.3%**
- Inflow of immigrants with **college education**:
  - Earnings gap b/w immigrants and natives **-5.9%** vs spatial earnings inequality **-0.1%**

Amenities estimates | Human capital estimates | Allocations | Competition vs. Skills Effect

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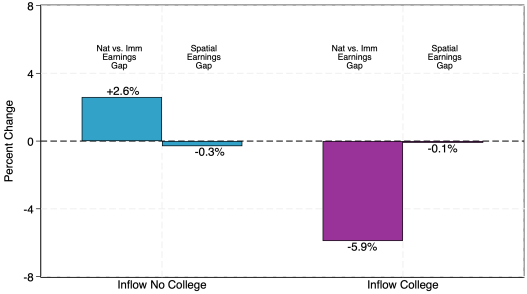

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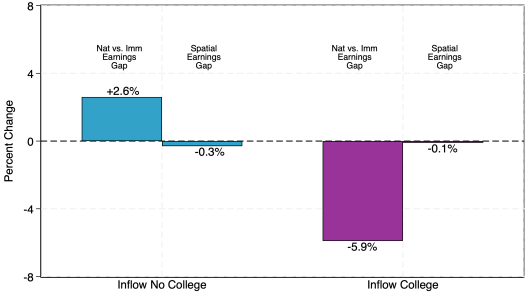
  - Earnings gap b/w immigrants and natives **-5.9%** vs spatial earnings inequality **-0.1%**

Amenities estimates | Human capital estimates | Allocations | Competition vs. Skills Effect

Immigration Policy

- GE responses after an inflow of immigrants (overall employment +1pp):
  - **Policy 1:** inflow of immigrants with no college education
  - **Policy 2:** inflow of immigrants with college education



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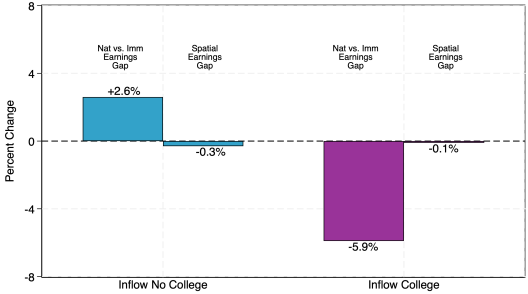

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Spatial Earnings Inequality ↓

Conclusion

Conclusion

What is the geography of immigrants' labor market outcomes in the US?

- Earnings gap with natives grows with city size only for immigrants from low-income countries
  - **Empirical contribution:** job choices across space depend on country of origin

How does it relate to earnings gaps with natives and spatial earnings inequality?

- **Model contribution:** rich spatial GE framework with occupational choices
- **Quantitative contribution:** study drivers of occupational choices across space
  i. No differences in **human capital** or **amenities** b/w immigrants and natives $\rightarrow$ **Inequality trade-off**
  ii. No origin-specific local labor market **wedges** $\rightarrow$ **No inequality trade-off**
    - Improved allocation of <u>all</u> workers into occupations across space

How does immigration policy affect these outcomes?

- How the earnings gap b/w immigrants and natives change depends on who enters the country
- **Spatial earnings inequality** $\downarrow$ regardless of who enters the country

Thank you!

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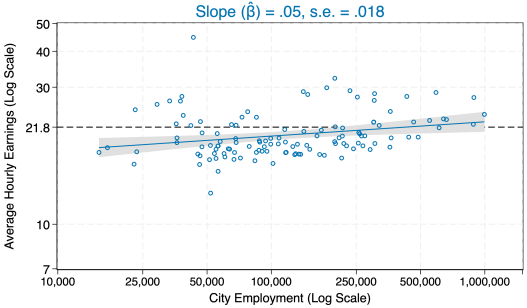
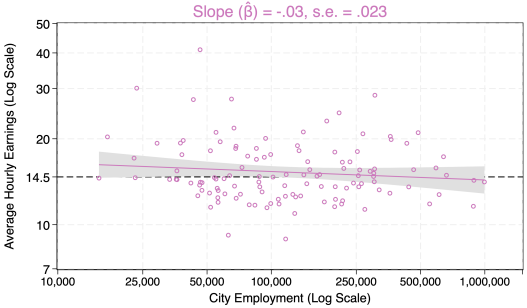
Data

- 2010 American Community Survey (ACS) sample from IPUMS:
  - Immigrants: foreign-born workers, first-generation
  - Hourly earnings
  - US cities: Metropolitan Statistical Areas (MSA)
  - Sample: male workers, 18-64 y.o., employed and work for wages
- O*NET:
  - Tasks intensity as in Acemoglu & Autor (2011)
- World Bank:
  - Countries GDP per capita 2017 USD
    - Low-income $\rightarrow$ GDP pc $<$ \$30,000
    - High-income $\rightarrow$ GDP pc $\geq$ \$30,000

City-Size Earnings Premia: Natives vs Immigrants



Natives

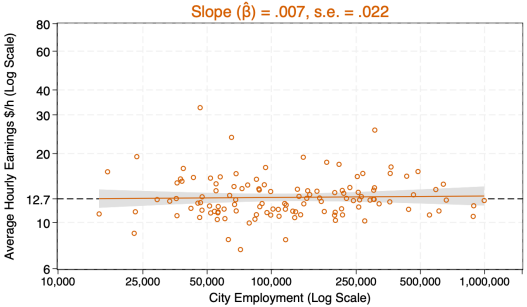


Immigrants

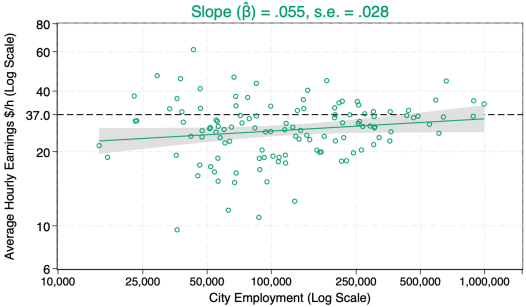
- The earnings gap b/w immigrants and natives is larger in big cities
  - Natives $\rightarrow$ doubling the city size increases hourly earnings by 3.6%
  - Immigrants $\rightarrow$ no significant change in earnings b/w small and big cities

Robustness 1 Male | Robustness 1 Male CP | Robustness 1 Male Conditional | Robustness 1 Male Conditional CP | Robustness Female 1 | Robustness Female 1 CP | Robustness 1 Female Conditional | Robustness 1 Female Conditional CP | Data & stylised facts

City-Size Earnings Premium by Country of Origin



Low-income



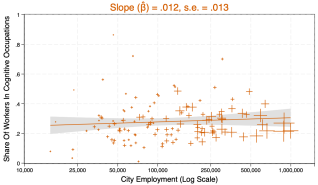
High-income

- No city-size earnings premia only for immigrants from low-income countries
  - High-income → doubling the city size increases hourly earnings by 3.9%

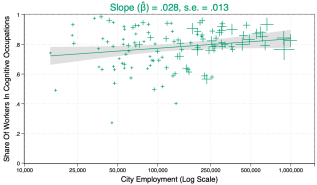
Robustness 2 Male | Robustness 2 Male CP | Robustness 2 Male Conditional | Robustness 2 Male Conditional CP | Robustness 2 Female | Robustness 2 Female CP | Robustness 2 Female Conditional | Robustness 2 Female Conditional CP | Table Natives vs Low-High Income | Stylised Facts

Spatial Distribution of Workers into Cognitive Occupations and Cities

Natives



Low-Income



High-Income

- High-income country workers choose more cognitive jobs, especially in large cities. Doubling the city size:
  - The share of natives in cognitive jobs +1pp
  - The share of immigrants from high-income countries in cognitive jobs +1.5pp
  - The share of immigrants from low-income countries does not change

Fact 3 Table Stylised Facts

Robustness Checks Fact 1

Econometric model: $\ln w_i = \alpha + \beta \ln \text{Employment}_{j(i)} + X_i + \varepsilon_i$

	Log Hourly Earnings (1)	**Log Hourly Earnings** (2)	**Log Hourly Earnings** (3)	**Log Hourly Earnings** (4)	**Log Hourly Earnings** (5)
			Immigrants		
Log City Employment	−0.049	−0.021	−0.024	−0.025	−0.014
	(0.021)	(0.011)	(0.012)	(0.014)	(0.012)
Constant	3.000	2.360	1.825	0.987	2.990
	(0.256)	(0.136)	(0.160)	(0.198)	(0.195)
N. Obs	56,999	56,999	56,999	56,999	56,999
Adj.R2	0.00	0.27	0.28	0.23	0.41
			Natives		
Log City Employment	0.068	0.039	0.046	0.049	0.042
	(0.013)	(0.008)	(0.008)	(0.008)	(0.007)
Constant	1.950	1.705	0.639	−0.646	1.720
	(0.155)	(0.095)	(0.102)	(0.105)	(0.096)
N. Obs	562,577	562,577	562,577	562,577	562,577
Adj.R2	0.01	0.23	0.35	0.34	0.45
Years of School FE	✗	✓	✓	✗	✓
Linear Years of School	✗	✗	✗	✓	✗
Experience FE	✗	✗	✓	✗	✓
Cubic Experience	✗	✗	✗	✓	✗
Occupation FE	✗	✗	✗	✗	✓
Origin FE	✗	✗	✗	✗	✗

Fact 1 Plots

Robustness Checks Fact 1 City Prices

	Log Hourly Earnings (1)	**Log Hourly Earnings** (2)	**Log Hourly Earnings** (3)	**Log Hourly Earnings** (4)	**Log Hourly Earnings** (5)
			Immigrants		
Log City Employment	−0.152	−0.126	−0.128	−0.130	−0.115
	(0.052)	(0.051)	(0.051)	(0.055)	(0.043)
Constant	−2.325	−2.922	−3.697	−4.287	−2.577
	(0.627)	(0.621)	(0.653)	(0.688)	(0.559)
N. Obs	56,999	56,999	56,999	56,999	56,999
Adj.R2	0.03	0.25	0.26	0.21	0.4
			Natives		
Log City Employment	−0.052	−0.079	−0.072	−0.069	−0.073
	(0.026)	(0.029)	(0.026)	(0.026)	(0.024)
Constant	−3.057	−3.332	−4.429	−5.572	−3.418
	(0.306)	(0.334)	(0.295)	(0.301)	(0.270)
N. Obs	562,577	562,577	562,577	562,577	562,577
Adj.R2	0.01	0.20	0.32	0.31	0.42
Years of School FE	✗	✓	✓	✗	✓
Linear Years of School	✗	✗	✗	✓	✗
Experience FE	✗	✗	✓	✗	✓
Cubic Experience	✗	✗	✗	✓	✗
Occupation FE	✗	✗	✗	✗	✓
Origin FE	✗	✗	✗	✗	✗

Fact 1 Plots

Robustness Checks Fact 1: Conditional Regressions

	No College Education	**College Education**	**0-14 Experience**	**15-29 Experience**	**30+ Experience**
	Log Hourly Earnings (1)	Log Hourly Earnings (2)	Log Hourly Earnings (3)	Log Hourly Earnings (4)	Log Hourly Earnings (5)
			Immigrants		
Log City Employment	−0.026	−0.030	−0.015	−0.031	−0.026
	(0.014)	(0.024)	(0.013)	(0.015)	(0.016)
Constant	2.302	3.333	2.151	2.567	2.612
	(0.176)	(0.310)	(0.168)	(0.189)	(0.195)
N. Obs	38,747	18,252	6,181	30,139	20,679
Adj.R2	0.03	0.01	0.36	0.23	0.12
			Natives		
Log City Employment	0.031	0.073	0.054	0.058	0.058
	(0.007)	(0.014)	(0.012)	(0.012)	(0.010)
Constant	1.777	1.840	1.500	1.852	1.950
	(0.090)	(0.170)	(0.143)	(0.144)	(0.124)
N. Obs	210,105	352,472	183,107	221,225	158,245
Adj.R2	0.13	0.08	0.17	0.16	0.12
College FE	✗	✗	✓	✓	✓
Experience FE	✓	✓	✗	✗	✗

Fact 1 Plots

Robustness Checks Fact 1: Conditional Regressions City Prices

	No College Education	**College Education**	**0-14 Experience**	**15-29 Experience**	**30+ Experience**
	Log Hourly Earnings (1)	Log Hourly Earnings (2)	Log Hourly Earnings (3)	Log Hourly Earnings (4)	Log Hourly Earnings (5)
			Immigrants		
Log City Employment	−0.137	−0.116	−0.133	−0.135	−0.124
	(0.049)	(0.059)	(0.05)	(0.051)	(0.052)
Constant	−2.911	−2.219	−2.966	−2.732	−2.802
	(0.593)	(0.726)	(0.609)	(0.621)	(0.633)
N. Obs	38,747	18,252	6,181	30,139	20,679
Adj.R2	0.06	0.03	0.34	0.23	0.12
			Natives		
Log City Employment	−0.087	−0.047	−0.073	−0.056	−0.055
	(0.026)	(0.024)	(0.026)	(0.024)	(0.025)
Constant	−3.246	−3.204	−3.414	−3.199	−3.112
	(0.313)	(0.285)	(0.307)	(0.281)	(0.291)
N. Obs	210,105	352,472	183,107	221,225	158,245
Adj.R2	0.14	0.09	0.15	0.14	0.10
College FE	✗	✗	✓	✓	✓
Experience FE	✓	✓	✗	✗	✗

Fact 1 Plots

Robustness Checks Fact 1: Female Workers

	Log Hourly Earnings (1)	**Log Hourly Earnings** (2)	**Log Hourly Earnings** (3)	**Log Hourly Earnings** (4)	**Log Hourly Earnings** (5)
			Immigrants		
Log City Employment	−0.015	−0.003	−0.004	0.000	−0.007
	(0.018)	(0.012)	(0.012)	(0.011)	(0.012)
Constant	2.363	1.941	1.689	0.884	2.861
	(0.222)	(0.149)	(0.186)	(0.169)	(0.263)
N. Obs	40,794	40,794	40,794	40,794	40,794
Adj.R2	0.00	0.22	0.22	0.19	0.38
			Natives		
Log City Employment	0.073	0.045	0.050	0.051	0.044
	(0.017)	(0.011)	(0.013)	(0.013)	(0.012)
Constant	1.670	1.438	0.587	−0.614	1.786
	(0.210)	(0.138)	(0.164)	(0.165)	(0.158)
N. Obs	479,097	479,097	479,097	479,097	479,097
Adj.R2	0.01	0.21	0.29	0.28	0.42
Years of School FE	✗	✓	✓	✗	✓
Linear Years of School	✗	✗	✗	✓	✗
Experience FE	✗	✗	✓	✗	✓
Cubic Experience	✗	✗	✗	✓	✗
Occupation FE	✗	✗	✗	✗	✓
Origin FE	✗	✗	✗	✗	✗

Fact 1 Plots

Robustness Checks Fact 1: Female Workers City Prices

	Log Hourly Earnings (1)	**Log Hourly Earnings** (2)	**Log Hourly Earnings** (3)	**Log Hourly Earnings** (4)	**Log Hourly Earnings** (5)
			Immigrants		
Log City Employment	−0.121	−0.110	−0.110	−0.106	−0.109
	(0.044)	(0.045)	(0.046)	(0.049)	(0.042)
Constant	−2.978	−3.369	−3.665	−4.466	−2.523
	(0.533)	(0.555)	(0.585)	(0.559)	(0.586)
N. Obs	40,794	40,794	40,794	40,794	40,794
Adj.R2	0.02	0.21	0.21	0.17	0.36
			Natives		
Log City Employment	−0.053	−0.078	−0.072	−0.072	−0.077
	(0.024)	(0.029)	(0.026)	(0.026)	(0.025)
Constant	−3.286	−3.547	−4.435	−5.491	−3.322
	(0.292)	(0.340)	(0.297)	(0.308)	(0.287)
N. Obs	479,097	479,097	479,097	479,097	479,097
Adj.R2	0.00	0.17	0.26	0.25	0.39
Years of School FE	✗	✓	✓	✗	✓
Linear Years of School	✗	✗	✗	✓	✗
Experience FE	✗	✗	✓	✗	✓
Cubic Experience	✗	✗	✗	✓	✗
Occupation FE	✗	✗	✗	✗	✓
Origin FE	✗	✗	✗	✗	✗

Fact 1 Plots

Robustness Checks Fact 1: Female Workers Conditional Regressions

	No College Education	**College Education**	**0-14 Experience**	**15-29 Experience**	**30+ Experience**
	Log Hourly Earnings (1)	Log Hourly Earnings (2)	Log Hourly Earnings (3)	Log Hourly Earnings (4)	Log Hourly Earnings (5)
			Immigrants		
Log City Employment	−0.020	0.025	0.003	0.003	−0.016
	(0.017)	(0.018)	(0.018)	(0.017)	(0.016)
Constant	2.109	2.285	1.819	1.939	2.261
	(0.202)	(0.252)	(0.229)	(0.203)	(0.201)
N. Obs	26,646	14,148	2,835	20,619	17,340
Adj.R2	0.01	0.00	0.24	0.17	0.13
			Natives		
Log City Employment	0.040	0.074	0.059	0.067	0.060
	(0.010)	(0.020)	(0.016)	(0.016)	(0.015)
Constant	01.533	01.675	01.296	01.508	01.668
	(0.124)	(0.239)	(0.193)	(0.202)	(0.185)
N. Obs	161,996	317,101	162,052	179,563	137,482
Adj.R2	0.08	0.04	0.17	0.14	0.11
College FE	✗	✗	✓	✓	✓
Experience FE	✓	✓	✗	✗	✗

Fact 1 Plots

Robustness Checks Fact 1: Female Workers Conditional Regressions City Prices

	No College Education	**College Education**	**0-14 Experience**	**15-29 Experience**	**30+ Experience**
	Log Hourly Earnings (1)	Log Hourly Earnings (2)	Log Hourly Earnings (3)	Log Hourly Earnings (4)	Log Hourly Earnings (5)
			Immigrants		
Log City Employment	−0.131	−0.070	−0.119	−0.103	−0.119
	(0.044)	(0.049)	(0.057)	(0.044)	(0.045)
Constant	−3.145	−3.191	−3.297	−3.386	−3.134
	(0.533)	(0.575)	(0.705)	(0.532)	(0.547)
N. Obs	26,646	14,148	2,835	20,619	17,340
Adj.R2	0.04	0.01	0.23	0.17	0.14
			Natives		
Log City Employment	−0.08	−0.053	−0.076	−0.052	−0.058
	(0.027)	(0.024)	(0.028)	(0.023)	(0.023)
Constant	−3.488	−3.294	−3.538	−3.511	−3.357
	(0.319)	(0.286)	(0.339)	(0.271)	(0.279)
N. Obs	161,996	317,101	162,052	179,563	137,482
Adj.R2	0.09	0.04	0.14	0.12	0.09
College FE	✗	✗	✓	✓	✓
Experience FE	✓	✓	✗	✗	✗

Fact 1 Plots

Robustness Checks Fact 2

	Log Hourly Earnings (1)	**Log Hourly Earnings** (2)	**Log Hourly Earnings** (3)	**Log Hourly Earnings** (4)	**Log Hourly Earnings** (5)
			Low-Income		
Log Employment	−0.039	−0.020	−0.024	−0.025	−0.016
	(0.018)	(0.012)	(0.012)	(0.014)	(0.011)
Constant	2.800	2.341	1.803	1.164	2.681
	(0.229)	(0.139)	(0.165)	(0.207)	(0.217)
N. Obs	51,470	51,470	51,470	51,470	51,470
Adj.R2	0.00	0.14	0.23	0.18	0.34
			High-Income		
Log Employment	0.059	0.052	0.063	0.067	0.048
	(0.027)	(0.020)	(0.020)	(0.022)	(0.016)
Constant	2.564	2.066	1.049	−0.917	2.127
	(0.346)	(0.289)	(0.321)	(0.355)	(0.378)
N. Obs	5,529	5,529	5,529	5,529	5,529
Adj.R2	0.00	0.29	0.24	0.2	0.38
			Natives		
Log Employment	0.068	0.039	0.046	0.049	0.042
	(0.013)	(0.008)	(0.008)	(0.008)	(0.007)
Constant	1.950	1.705	0.639	−0.646	1.720
	(0.155)	(0.095)	(0.102)	(0.105)	(0.096)
N. Obs	562,577	562,577	562,577	562,577	562,577
Adj.R2	0.01	0.09	0.35	0.34	0.45
Years of School FE	✗	✓	✓	✗	✓
Linear Years of School	✗	✗	✗	✓	✗
Experience FE	✗	✗	✓	✗	✓
Cubic Experience	✗	✗	✗	✓	✗
Occupation FE	✗	✗	✗	✗	✓

Fact 2 Plots

Robustness Checks Fact 2 City Prices

	Log Hourly Earnings (1)	**Log Hourly Earnings** (2)	**Log Hourly Earnings** (3)	**Log Hourly Earnings** (4)	**Log Hourly Earnings** (5)
			Low-Income		
Log Employment	−0.143	−0.125	−0.128	−0.129	−0.116
	(0.053)	(0.052)	(0.053)	(0.056)	(0.044)
Constant	−2.522	−2.939	−3.797	−4.106	−2.981
	(0.641)	(0.635)	(0.733)	(0.699)	(0.671)
N. Obs	51,470	51,470	51,470	51,470	51,470
Adj.R2	0.03	0.64	0.21	0.16	0.34
			High-Income		
Log Employment	−0.044	−0.050	−0.038	−0.035	−0.048
	(0.059)	(0.05)	(0.046)	(0.047)	(0.040)
Constant	−2.773	−3.386	−4.592	−6.366	−3.421
	(0.710)	(0.564)	(0.635)	(0.675)	(0.682)
N. Obs	5,529	5,529	5,529	5,529	5,529
Adj.R2	0.00	0.56	0.23	0.19	0.37
			Natives		
Log Employment	−0.052	−0.079	−0.072	−0.069	−0.073
	(0.026)	(0.029)	(0.026)	(0.026)	(0.024)
Constant	−3.057	−3.332	−4.429	−5.572	−3.418
	(0.306)	(0.334)	(0.295)	(0.301)	(0.270)
N. Obs	562,577	562,577	562,577	562,577	562,577
Adj.R2	0.00	0.33	0.32	0.31	0.42
Years of School FE	✗	✓	✓	✗	✓
Linear Years of School	✗	✗	✗	✓	✗
Experience FE	✗	✗	✓	✗	✓
Cubic Experience	✗	✗	✗	✓	✗
Occupation FE	✗	✗	✗	✗	✓

Fact 2 Plots

Robustness Checks Fact 2: Conditional Regressions

	No College Education	**College Education**	**0-14 Experience**	**15-29 Experience**	**30+ Experience**
	Log Hourly Earnings (1)	Log Hourly Earnings (2)	Log Hourly Earnings (3)	Log Hourly Earnings (4)	Log Hourly Earnings (5)
			Low-Income		
Log City Employment	−0.023	−0.035	−0.025	−0.030	−0.019
	(0.014)	(0.025)	(0.013)	(0.016)	(0.014)
Constant	02.251	03.283	02.277	02.544	02.499
	(0.170)	(0.317)	(0.173)	(0.198)	(0.173)
N. Obs	37,308	14,162	5,568	27,059	18,843
Adj.R2	0.03	0.01	0.3	0.17	0.08
			High-Income		
Log City Employment	0.030	0.081	0.082	0.054	0.087
	(0.026)	(0.032)	(0.046)	(0.025)	(0.037)
Constant	2.274	2.237	1.625	2.111	1.724
	(0.353)	(0.406)	(0.597)	(0.327)	(0.459)
N. Obs	1,439	4,090	613	3,080	1,836
Adj.R2	0.00	0.03	0.10	0.17	0.17
			Natives		
Log City Employment	0.031	0.073	0.054	0.058	0.058
	(0.007)	(0.014)	(0.012)	(0.012)	(0.01)
Constant	1.777	1.840	1.500	1.852	1.950
	(0.090)	(0.170)	(0.143)	(0.144)	(0.124)
N. Obs	210,105	352,472	183,107	221,225	158,245
Adj.R2	0.13	0.08	0.17	0.16	0.12
College FE	✗	✗	✓	✓	✓
Experience FE	✓	✓	✗	✗	✗

Fact 2 Plots

Robustness Checks Fact 2: Conditional Regressions City Prices

	No College Education	**College Education**	**0-14 Experience**	**15-29 Experience**	**30+ Experience**
	Log Hourly Earnings (1)	Log Hourly Earnings (2)	Log Hourly Earnings (3)	Log Hourly Earnings (4)	Log Hourly Earnings (5)
			Low-Income		
Log City Employment	−0.133	−0.117	−0.141	−0.135	−0.117
	(0.050)	(0.065)	(0.056)	(0.054)	(0.051)
Constant	−2.967	−2.288	−2.870	−2.744	−2.921
	(0.603)	(0.771)	(0.683)	(0.652)	(0.621)
N. Obs	37,308	14,162	5,568	27,059	18,843
Adj.R2	0.06	0.03	0.30	0.17	0.09
			High-Income		
Log City Employment	−0.106	−0.009	−0.058	−0.043	−0.009
	(0.068)	(0.041)	(0.043)	(0.046)	(0.057)
Constant	−2.643	−3.321	−3.254	−3.313	−3.739
	(0.849)	(0.514)	(0.557)	(0.529)	(0.685)
N. Obs	1,439	4,090	613	3,080	1,836
Adj.R2	0.02	0.03	0.08	0.17	0.18
			Natives		
Log City Employment	−0.087	−0.047	−0.073	−0.056	−0.055
	(0.026)	(0.024)	(0.026)	(0.024)	(0.025)
Constant	−3.246	−3.204	−3.414	−3.199	0 − 3.112
	(0.313)	(0.285)	(0.307)	(0.281)	(0.291)
N. Obs	210,105	352,472	183,107	221,225	158,245
Adj.R2	0.14	0.09	0.15	0.14	0.1
College FE	✗	✗	✓	✓	✓
Experience FE	✓	✓	✗	✗	✗

Fact 2 Plots

Robustness Checks Fact 2: Female Workers

	Log Hourly Earnings (1)	**Log Hourly Earnings** (2)	**Log Hourly Earnings** (3)	**Log Hourly Earnings** (4)	**Log Hourly Earnings** (5)
			Low-Income		
Log Employment	−0.009	0.001	−0.001	0.003	−0.007
	(0.017)	(0.012)	(0.012)	(0.011)	(0.012)
Constant	2.253	1.890	1.644	0.853	2.577
	(0.214)	(0.148)	(0.190)	(0.169)	(0.312)
N. Obs	37,531	37,531	37,531	37,531	37,531
Adj.R2	0.00	0.15	0.20	0.17	0.35
			High-Income		
Log Employment	0.053	0.018	0.027	0.028	0.021
	(0.032)	(0.027)	(0.028)	(0.029)	(0.025)
Constant	2.040	1.925	0.556	−0.080	1.496
	(0.406)	(0.343)	(0.543)	(0.534)	(0.665)
N. Obs	3,263	3,263	3,263	3,263	3,263
Adj.R2	0.00	0.34	0.22	0.19	0.40
			Natives		
Log Employment	0.073	0.045	0.050	0.051	0.044
	(0.017)	(0.011)	(0.013)	(0.013)	(0.012)
Constant	1.670	1.438	0.587	−0.614	1.786
	(0.21)	(0.138)	(0.164)	(0.165)	(0.158)
N. Obs	479,097	479,097	479,097	479,097	479,097
Adj.R2	0.01	0.14	0.29	0.28	0.42
Years of School FE	✗	✓	✓	✗	✓
Linear Years of School	✗	✗	✗	✓	✗
Experience FE	✗	✗	✓	✗	✓
Cubic Experience	✗	✗	✗	✓	✗
Occupation FE	✗	✗	✗	✗	✓

Fact 2 Plots

Robustness Checks Fact 2: Female Workers City Prices

	Log Hourly Earnings (1)	**Log Hourly Earnings** (2)	**Log Hourly Earnings** (3)	**Log Hourly Earnings** (4)	**Log Hourly Earnings** (5)
			Low-Income		
Log Employment	−0.114	−0.105	−0.106	−0.102	−0.108
	(0.044)	(0.046)	(0.046)	(0.049)	(0.043)
Constant	−3.11	−3.439	−3.727	−4.509	−2.893
	(0.536)	(0.558)	(0.589)	(0.565)	(0.594)
N. Obs	37,531	37,531	37,531	37,531	37,531
Adj.R2	0.02	0.56	0.19	0.15	0.33
			High-Income		
Log Employment	−0.065	−0.096	−0.086	−0.087	−0.085
	(0.055)	(0.048)	(0.044)	(0.047)	(0.034)
Constant	−3.116	−3.345	−4.507	−5.364	−3.536
	(0.666)	(0.577)	(0.694)	(0.594)	(0.65)
N. Obs	3,263	3,263	3,263	3,263	3,263
Adj.R2	0.01	0.58	0.21	0.18	0.40
			Natives		
Log Employment	−0.053	−0.078	−0.072	−0.072	−0.077
	(0.024)	(0.029)	(0.026)	(0.026)	(0.025)
Constant	−3.286	−3.547	−4.435	−5.491	−3.322
	(0.292)	(0.340)	(0.297)	(0.308)	(0.287)
N. Obs	479,097	479,097	479,097	479,097	479,097
Adj.R2	0.00	0.34	0.26	0.25	0.39
Years of School FE	✗	✓	✓	✗	✓
Linear Years of School	✗	✗	✗	✓	✗
Experience FE	✗	✗	✓	✗	✓
Cubic Experience	✗	✗	✗	✓	✗
Occupation FE	✗	✗	✗	✗	✓

Fact 2 Plots

Robustness Checks Fact 2: Female Workers Conditional Regressions

	No College Education	**College Education**	**0-14 Experience**	**15-29 Experience**	**30+ Experience**
	Log Hourly Earnings (1)	Log Hourly Earnings (2)	Log Hourly Earnings (3)	Log Hourly Earnings (4)	Log Hourly Earnings (5)
			Low-Income		
Log City Employment	−0.016	0.031	0.001	0.004	−0.009
	(0.016)	(0.018)	(0.020)	(0.016)	(0.016)
Constant	2.048	2.12	1.826	1.917	2.160
	(0.201)	(0.247)	(0.252)	(0.194)	(0.199)
N. Obs	25,450	12,081	2,520	18,995	16,016
Adj.R2	0.01	00	0.2	0.15	0.12
			High-Income		
Log City Employment	0.019	0.057	0.000	0.107	−0.023
	(0.030)	(0.045)	(0.055)	(0.04)	(0.042)
Constant	2.076	2.213	2.318	1.072	2.634
	(0.406)	(0.572)	(0.719)	(0.502)	(0.536)
N. Obs	1,196	2,067	315	1,624	1,324
Adj.R2	0.00	0.01	0.13	0.13	0.13
			Natives		
Log City Employment	0.040	0.074	0.059	0.067	0.060
	(0.010)	(0.020)	(0.016)	(0.016)	(0.015)
Constant	1.533	1.675	1.296	1.508	1.668
	(0.124)	(0.239)	(0.193)	(0.202)	(0.185)
N. Obs	161,996	317,101	162,052	179,563	137,482
Adj.R2	0.08	0.04	0.17	0.14	0.11
College FE	✗	✗	✓	✓	✓
Experience FE	✓	✓	✗	✗	✗

Fact 2 Plots

Robustness Checks Fact 2: Female Workers Conditional Regressions City Prices

	No College Education	**College Education**	**0-14 Experience**	**15-29 Experience**	**30+ Experience**
	Log Hourly Earnings (1)	Log Hourly Earnings (2)	Log Hourly Earnings (3)	Log Hourly Earnings (4)	Log Hourly Earnings (5)
			Low-Income		
Log City Employment	−0.126	−0.061	−0.119	−0.101	−0.110
	(0.044)	(0.050)	(0.061)	(0.045)	(0.044)
Constant	−3.222	−3.386	−3.309	−3.417	−3.270
	(0.542)	(0.576)	(0.746)	(0.546)	(0.540)
N. Obs	25,450	12,081	2,520	18,995	16,016
Adj.R2	0.04	0.01	0.20	0.15	0.12
			High-Income		
Log City Employment	−0.120	−0.045	−0.136	−0.004	−0.142
	(0.055)	(0.051)	(0.073)	(0.053)	(0.056)
Constant	−2.880	−3.204	−2.675	−4.191	−2.522
	(0.670)	(0.634)	(0.912)	(0.649)	(0.681)
N. Obs	1,196	2,067	315	1,624	1,324
Adj.R2	0.03	0.00	0.15	0.11	0.17
			Natives		
Log City Employment	−0.080	−0.053	−0.076	−0.052	−0.058
	(0.027)	(0.024)	(0.028)	(0.023)	(0.023)
Constant	−3.488	−3.294	−3.538	−3.511	−3.357
	(0.319)	(0.286)	(0.339)	(0.271)	(0.279)
N. Obs	161,996	317,101	162,052	179,563	137,482
Adj.R2	0.09	0.04	0.14	0.12	0.09
College FE	✗	✗	✓	✓	✓
Experience FE	✓	✓	✗	✗	✗

Fact 2 Plots

Hourly Earnings: Big vs Small Cities

	Small City (Pop. $<$ 500,000)	Big City (Pop. $\geq$ 500,000)	City-Size Gap
Natives	21.0	23.8	+2.8
High-Income	33.2	39.6	+6.4
Low-Income	13.3	11.9	-1.4

Fact 2 Plots

Workers Distributions across Cities and Occupations

		Small City (Pop. $<$ 500,000)	Big City (Pop. $\geq$ 500,000)	Δ
Natives	% Cognitive	63.9	68.8	4.9
	% Total	17.7	82.3	64.6
High-Income	% Cognitive	71.6	80.4	8.9
	% Total	19.3	80.7	61.3
Low-Income	% Cognitive	27.5	24.7	-2.8
	% Total	10.7	89.3	78.7

- Workers from high-income countries work more in cognitive jobs in big cities
- Workers from low-income countries are more likely to live in big cities relative to all other workers

The Problem of the Firm

- A firm in city j solves:

$$\max Y_j = \left[M_j^{\frac{\sigma-1}{\sigma}} + (\theta_j D_j)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} - r_{jD} D_j - r_{jM} M_j$$

where:

- σ is the elasticity of substitution between the two inputs
- r_{jo} is the city-occupation-specific skills price

- The city-occupation-specific skills price ratio is:

$$\frac{r_{jD}}{r_{jM}} = \left(\frac{D_j}{M_j} \right)^{-\frac{1}{\sigma}} \theta_j^{\left(1-\frac{1}{\sigma}\right)}$$

The Problem of the Worker and Demands for Goods

- Given their city-occupation choice, a worker i from group g solves:

$$\max_{c_{jog}, h_{jog}} \quad U_{jog} = c_{jog}^{(1-\alpha)} h_{jog}^{\alpha} z_{jog} \exp\{\varepsilon_{jo}\}$$

$$\text{s.t.} \quad c_{jog} + p_j h_{jog} \leq w_{jog}$$

where

  - c consumption good, h housing good, α expenditure share in the housing good

- Demands for goods are:

$$c_{jog}^{\star} = (1-\alpha)\, w_{jog}$$

$$h_{jog}^{\star} = \alpha \frac{w_{jog}}{p_j}$$

The Problem of the Worker and Demands for Goods

- Given their city-occupation choice, a worker i from group g solves:

$$\max_{c_{jog}, h_{jog}} \quad U_{jog} = c_{jog}^{(1-\alpha)} h_{jog}^{\alpha} z_{jog} \exp\{\varepsilon_{jo}\}$$

$$\text{s.t.} \quad c_{jog} + p_j h_{jog} \leq w_{jog}$$

where

  - c consumption good, h housing good, α expenditure share in the housing good

- Demands for goods are:

$$c_{jog}^{\star} = (1-\alpha)\, w_{jog}$$

$$h_{jog}^{\star} = \alpha \frac{w_{jog}}{p_j}$$

Indirect Utility and Choice Equation

- Indirect utility from living in city j and working in occupation o is:

$$V_{jog} = \gamma p_j^{-\alpha} w_{jog} z_{jog} \exp\{\varepsilon_{jo}\}$$

where $\gamma = (1-\alpha)^{(1-\alpha)}\,\alpha^{\alpha}$

- The share of workers from group g choosing a city j and an occupation o is:

$$\pi_{jog} = \frac{\gamma p_j^{-\alpha} w_{jog} z_{jog}}{\sum_{j' \in \mathcal{J}} \sum_{o' \in \mathcal{O}} \gamma p_{j'}^{-\alpha} w_{j'o'g} z_{j'o'g}}$$

$$= \frac{\gamma p_j^{-\alpha} r_{jo} s_{og} \tau_{jog} z_{jog}}{\sum_{j' \in \mathcal{J}} \sum_{o' \in \mathcal{O}} \gamma p_{j'}^{-\alpha} r_{j'o'} s_{o'g'} \tau_{j'o'g'} z_{j'o'g}}$$

Endogenous Housing Supply

- The production function for housing is given by:

$$H_j = f(Y_j, T_j) = \omega_j Y_j^{\iota_j} T_j^{1-\iota_j}$$

where $\omega_j = \iota_j^{-\iota_j}$ is a constant, and $(1 - \iota_j)$ is the weight of land in the production of housing.

- The (absentee) landlord solves:

$$\max_{Y_j} \quad p_j \left(\omega_j Y_j^{\iota_j} T_j^{1-\iota_j} \right) - Y_j$$

- Solving FOC and rearranging:

$$Y_j = \left(p_j \omega_j \iota_j \right)^{\frac{1}{1-\iota}} T_j$$

- Plug FOC into the production function to get the housing supply in a city j:

$$p_j = \left(\frac{H_j}{T_j} \right)^{\frac{1}{\zeta_j}}$$

Spatial Equilibrium

- A spatial equilibrium is a set of skills prices $\{r^\star_{jo}\}_{j\in\mathcal{J},o\in\mathcal{O}}$, housing prices $\{p^\star_j\}_{j\in\mathcal{J}}$, an allocation of workers across locations and occupations $\{\pi^\star_{jog}\}_{j\in\mathcal{J},o\in\mathcal{O}}$, such that:
  - The share of workers from group g in a city-occupation pair jo is:

$$\pi^\star_{jog} = \frac{\gamma p_j^{\star\,-\alpha} r^\star_{jo} s_{og} \tau_{jog} z_{jog}}{\sum_{j'\in\mathcal{J}}\sum_{o'\in\mathcal{O}} \gamma p_{j'}^{\star\,-\alpha} r^\star_{j'o'} s_{o'g} \tau_{j'o'g} z_{j'o'g}}$$

  - Labor supply satisfies:

$$M^\star_j = \sum_g \pi^\star_{jMg} s_{Mg} \phi_g, \quad D^\star_j = \sum_g \pi^\star_{jDg} s_{Dg} \phi_g$$

  - Labor markets clear for each city-occupation pair, that is $\forall j \in \mathcal{J}$:

$$r^\star_{jM} = \frac{\left[M_j^{\star\,\frac{\sigma-1}{\sigma}} + (\theta_j D^\star_j)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{1}{\sigma-1}}}{M_j^{\star\,\frac{1}{\sigma}}}, \quad r^\star_{jD} = \frac{\left[M_j^{\star\,\frac{\sigma-1}{\sigma}} + (\theta_j D^\star_j)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{1}{\sigma-1}}}{D_j^{\star\,\frac{1}{\sigma}}} \theta_j^{\left(1-\frac{1}{\sigma}\right)}$$

  - The housing market clear in each city, that is $\forall j \in \mathcal{J}$:

$$p^\star_j = \left[\frac{\alpha}{T_j}\sum_o\sum_g \pi^\star_{jog}\phi_g r^\star_{jo} s_{og}\tau_{jog}\right]^{\frac{1}{\varsigma_j-1}}$$

From the Model to the Data: Internally Calibrated Parameters & Identification

Parameters Calibrated Using MSM

	Description	N. Parameters	Value
θ_j	City productivity bias	2	Bias
s_{og}	Human capital	36	Human capital
τ_{jok}	Wedge on earnings	8	Wedge on earnings
z_{jog}	Amenities	54	Amenities

Targeted Moments

Moment	N. Moments
Avg. natives earnings in city j and cognitive occupation	2
Avg. earnings in occupation o, $\forall g, o$	36
Avg. earnings for country of origin k in city j, occupation o, $\forall k \in \{\text{Low}, \text{High}\}, j, o$	8
Share of workers in group g in city j and occupation o	54

Model Fit: Earnings Model Fit: Shares Model Fit: Granular HC Model Fit: Granular Earnings Model Fit: Granular Shares Identification

Externally Calibrated Parameters

Parameters From The Literature Or Assumed

Description	Symbol	Value	Source
Elasticity of substitution	σ	3	Hsieh et al. (2019)
Housing supply elasticity	ζ	1.54	Saiz (2010)
Share of expenditure in housing	α	0.32	Albouy (2008)
Share of group g in the economy	ϕ		ACS 2010
Small & Big City Land	T	1	Assumed

Estimated City Productivity Bias In Cognitive Occupations

	Small City (1)	Big City (2)
Productivity Bias In Cognitive Occupations	1.3	1.5

Parameters

Estimated Human Capital

Workers Origins	Non-Cognitive Occupation (1)	Cognitive Occupation (2)	Overall (3)
Natives	7.0 (1.3)	15.2 (5.6)	11.1 (5.8)
High-Income	7.1 (0.9)	22.5 (6.0)	14.8 (8.9)
Low-Income	4.6 (0.7)	11.6 (4.4)	8.1 (4.7)

Parameters

Estimated Wedges on Earnings

Workers Origins	Small City		Big City	
	Non-Cognitive Occupation (1)	Cognitive Occupation (2)	Non-Cognitive Occupation (3)	Cognitive Occupation (4)
High-Income	1.3	1.1	1.2	1.1
Low-Income	1.2	0.9	1.0	0.7

Parameters

Estimated Amenities

Workers Origins	Small City		Big City	
	Non-Cognitive Occupation (1)	Cognitive Occupation (2)	Non-Cognitive Occupation (3)	Cognitive Occupation (4)
	Amenities			
Natives	1.0 (0.0)	1.3 (0.8)	3.9 (0.2)	6.4 (4.5)
High-Income	1.0 (0.0)	1.3 (1.1)	3.2 (1.4)	7.1 (7.7)
Low-Income	1.0 (0.0)	0.5 (0.4)	9.5 (2.2)	4.7 (3.6)

Parameters

Model Fit: Earnings

	Small City (Pop. $<$ 500,000)		Big City (Pop. $\geq$ 500,000)		Δ	
	Data (1)	Model (2)	Data (3)	Model (4)	Data (5)	Model (6)
Natives	21.0	20.6	23.8	23.6	+2.8	+3.0
High-Income	33.2	33.3	39.6	40.0	+6.4	+6.7
Low-Income	13.3	13.7	11.9	12.1	-1.4	-1.6

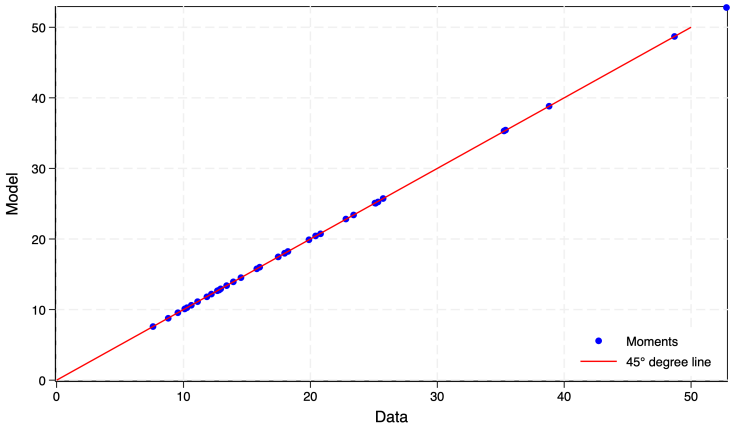
Parameters

Model Fit: Shares

		Small City (Pop. $<$ 500,000)		Big City (Pop. $\geq$ 500,000)		Δ	
		Data (1)	Model (2)	Data (3)	Model (4)	Data (5)	Model (6)
Natives	Cognitive Occ.	63.9	62.2	68.8	67.8	4.9	5.6
	Employment	17.7	18.0	82.3	82.0	64.6	64.1
High-Income	Cognitive Occ.	71.6	71.5	80.4	81.3	8.9	9.8
	Employment	19.3	17.2	80.7	82.8	61.3	65.6
Low-Income	Cognitive Occ.	27.5	29.6	24.7	25.8	-2.8	-3.8
	Employment	10.7	10.0	89.3	90.0	78.7	80.0

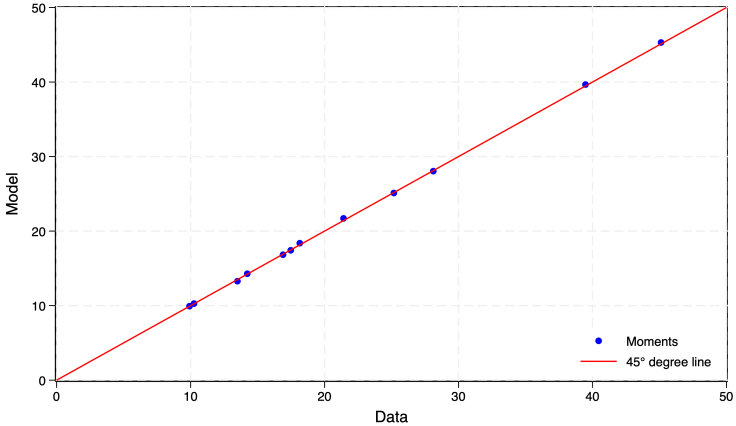
Parameters

Model Fit: Granular Human Capital



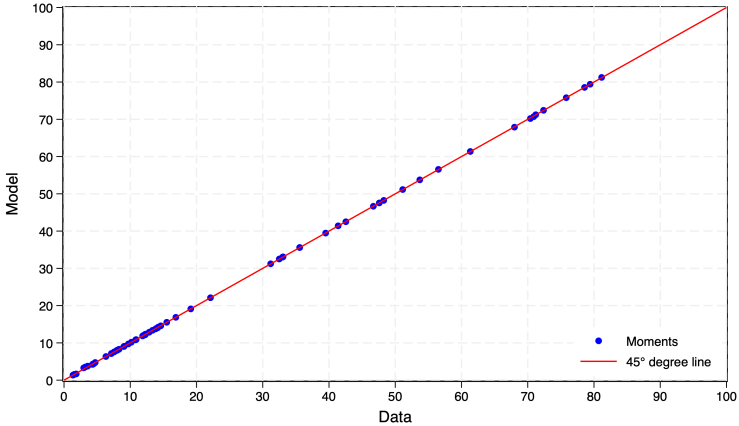
Parameters

Model Fit: Granular Earnings



Parameters

Model Fit: Granular Shares



Parameters

Real Output pc & Housing Prices Under The 5 Counterfactuals

- I measure the earnings gap between natives and immigrants as the ratio of the average natives' and immigrants' earnings:

$$\overline{W}^{\text{Gap}}_{\text{Workers}} = \frac{\overline{W}_{\text{US}}}{\overline{W}_{\text{Imm}}} = \frac{\sum_j \sum_o \sum_e \sum_x \pi_{joUSex} \phi_{USex} W_{joUSex}}{\sum_j \sum_o \sum_{k \neq \text{US}} \sum_e \sum_x \pi_{jokex} \phi_{kex} W_{jokex}}$$

- Similarly, I define spatial earnings inequality as the ratio of average earnings in the big city and in the small city:

$$\overline{W}^{\text{Gap}}_{\text{Cities}} = \frac{\overline{W}_{\text{Big}}}{\overline{W}_{\text{Small}}} = \frac{\sum_o \sum_k \sum_e \sum_x \pi_{\text{Big}okex} \phi_{kex} W_{\text{Big}okex}}{\sum_o \sum_k \sum_e \sum_x \pi_{\text{Small}okex} \phi_{kex} W_{\text{Small}okex}}$$

Main result

Earnings Gaps Under The 5 Counterfactuals

	Baseline	Counterfactuals				
		Same Human Capital As Natives	Same Amenities As Natives	No Wedges On Earnings	Same Amenities As Natives & No Wedges On Earnings	Full
		(1)	(2)	(3)	(4)	(5)
Parameters						
$s_{okex} = s_{oUSex}$	-	x	-	-	-	x
$z_{jokex} = z_{joUSex}$	-	-	x	-	x	x
$\tau_{jok} = 1$	-	-	-	x	x	x
$\overline{w}^{Gap}_{Workers}$	1	**0.811**	**0.938**	**0.907**	0.813	0.710
$\overline{w}^{Gap}_{Cities}$	1	**1.011**	**1.030**	**0.999**	1.025	1.023

Main result

Earnings Gaps Under The 5 Counterfactuals

	Baseline	Counterfactuals				
		Same Human Capital As Natives (1)	Same Amenities As Natives (2)	No Wedges On Earnings (3)	Same Amenities As Natives & No Wedges On Earnings (4)	Full (5)
Parameters						
$s_{okex} = s_{oUSex}$	-	x	-	-	-	x
$z_{jokex} = z_{joUSex}$	-	-	x	-	x	x
$\tau_{jok} = 1$	-	-	-	x	x	x
$\overline{w}^{Gap}_{Workers}$	1	**0.811**	**0.938**	**0.907**	0.813	0.710
$\overline{w}^{Gap}_{Cities}$	1	**1.011**	**1.030**	**0.999**	1.025	1.023

Main result

Earnings Gaps Under The 5 Counterfactuals

	Baseline	Counterfactuals				
		Same Human Capital As Natives (1)	Same Amenities As Natives (2)	No Wedges On Earnings (3)	Same Amenities As Natives & No Wedges On Earnings (4)	Full (5)
Parameters						
$s_{okex} = s_{oUSex}$	-	x	-	-	-	x
$z_{jokex} = z_{joUSex}$	-	-	x	-	x	x
$\tau_{jok} = 1$	-	-	-	x	x	x
$\overline{w}^{\text{Gap}}_{\text{Workers}}$	1	**0.811**	**0.938**	**0.907**	0.813	0.710
$\overline{w}^{\text{Gap}}_{\text{Cities}}$	1	**1.011**	**1.030**	**0.999**	1.025	1.023

Main result

Earnings Gaps Under The 5 Counterfactuals

	Baseline	Counterfactuals				
		Same Human Capital As Natives	Same Amenities As Natives	No Wedges On Earnings	Same Amenities As Natives & No Wedges On Earnings	Full
		(1)	(2)	(3)	(4)	(5)
Parameters						
$s_{okex} = s_{oUSex}$	-	x	-	-	-	x
$z_{jokex} = z_{joUSex}$	-	-	x	-	x	x
$\tau_{jok} = 1$	-	-	-	x	x	x
$\overline{w}^{Gap}_{Workers}$	1	**0.811**	**0.938**	**0.907**	0.813	0.710
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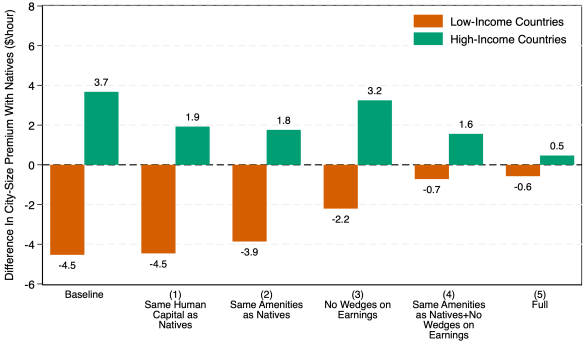
Main result

Earnings Gaps Under The 5 Counterfactuals

	Baseline	Counterfactuals				
		Same Human Capital As Natives	Same Amenities As Natives	No Wedges On Earnings	Same Amenities As Natives & No Wedges On Earnings	Full
		(1)	(2)	(3)	(4)	(5)
Parameters						
$s_{okex} = s_{oUSex}$	-	x	-	-	-	x
$z_{jokex} = z_{joUSex}$	-	-	x	-	x	x
$\tau_{jok} = 1$	-	-	-	x	x	x
$\overline{w}^{\text{Gap}}_{\text{Workers}}$	1	**0.811**	**0.938**	**0.907**	0.813	0.710
$\overline{w}^{\text{Gap}}_{\text{Cities}}$	1	**1.011**	**1.030**	**0.999**	1.025	1.023

Main result

What Determines the Differences in City-Size Earnings Premia?



Differences in city-size earnings premium:

- Human capital → no changes for low-income, -48.6% high-income
- Amenities → +13.3% low-income, -51.3% high-income
- Wedges on earnings → +51.1% low-income, -13.5% high-income
- Amenities & wedges → +84.4% low-income, -56.7% high-income Main result

Real Output pc & Housing Prices Under The 5 Counterfactuals

	Baseline	Counterfactuals				
		Same Human Capital As Natives	Same Amenities As Natives	No Wedges On Earnings	Same Amenities As Natives & No Wedge On Earnings	Full
		(1)	(2)	(3)	(4)	(5)
Parameters						
$s_{okex} = s_{oUSex}$	-	x	-	-	-	x
$z_{jokex} = z_{joUSex}$	-	-	x	-	x	x
$\tau_{jok} = 1$	-	-	-	x	x	x
			Housing Prices			
Big-Small City Ratio	1	1.010	1.026	1.008	1.034	1.031
			Real Output Per Capita			
US	1	1.018	1.007	1.002	1.009	1.023

- No differences in human capital:
  - Output gains larger than spatial increase in housing prices
- No differences in amenities or no wedges on earnings:
  - Output gains less than spatial increase in housing prices

Main result

Real Output pc & Housing Prices Under The 5 Counterfactuals

	Baseline	Counterfactuals				
		Same Human Capital As Natives	Same Amenities As Natives	No Wedges On Earnings	Same Amenities As Natives & No Wedge On Earnings	Full
		(1)	(2)	(3)	(4)	(5)
Parameters						
$s_{okex} = s_{oUSex}$	-	x	-	-	-	x
$z_{jokex} = z_{joUSex}$	-	-	x	-	x	x
$\tau_{jok} = 1$	-	-	-	x	x	x
			Housing Prices			
Big-Small City Ratio	1	1.010	1.026	1.008	1.034	1.031
			Real Output Per Capita			
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Main result

Real Output pc & Housing Prices Under The 5 Counterfactuals

	Baseline	Counterfactuals				
		Same Human Capital As Natives	Same Amenities As Natives	No Wedges On Earnings	Same Amenities As Natives & No Wedge On Earnings	Full
		(1)	(2)	(3)	(4)	(5)
Parameters						
$s_{ojkex} = s_{oUSex}$	-	x	-	-	-	x
$z_{jojkex} = z_{jojUSex}$	-	-	x	-	x	x
$\tau_{jojk} = 1$	-	-	-	x	x	x
			Housing Prices			
Big-Small City Ratio	1	1.010	1.026	1.008	1.034	1.031
			Real Output Per Capita			
US	1	1.018	1.007	1.002	1.009	1.023

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Main result

Real Output pc & Housing Prices Under The 5 Counterfactuals

	Baseline	Counterfactuals				
		Same Human Capital As Natives	Same Amenities As Natives	No Wedges On Earnings	Same Amenities As Natives & No Wedge On Earnings	Full
		(1)	(2)	(3)	(4)	(5)
Parameters						
$s_{okex} = s_{oUSex}$	-	x	-	-	-	x
$z_{jokex} = z_{joUSex}$	-	-	x	-	x	x
$\tau_{jok} = 1$	-	-	-	x	x	x
			Housing Prices			
Big-Small City Ratio	1	1.010	1.026	1.008	1.034	1.031
			Real Output Per Capita			
US	1	1.018	1.007	1.002	1.009	1.023

- No differences in human capital:
  - Output gains larger than spatial increase in housing prices
- No differences in amenities or no wedges on earnings:
  - Output gains less than spatial increase in housing prices

Main result

Mechanism: Workers' Reallocation across Cities

	Baseline	Counterfactuals				
		Same Human Capital As Natives	Same Amenities As Natives	No Wedges On Earnings	Same Amenities As Natives & No Wedges On Earnings	Full
		(1)	(2)	(3)	(4)	(5)
Parameters						
$s_{okex} = s_{oUSex}$	-	x	-	-	-	x
$z_{jokex} = z_{joUSex}$	-	-	x	-	x	x
$\tau_{jok} = 1$	-	-	-	x	x	x
		Share Of Workers In The Big City				
Natives	82.0	-0.2	-0.4	-0.1	-0.5	-0.4
High-Income	82.8	-0.6	-1.5	0.5	-1.0	-1.1
Low-Income	90.0	-0.1	-12.3	1.2	-9.5	-9.6

- Big-to-small cities reallocation:
  - No differences in human capital $\rightarrow$ workers from all countries
  - No differences in amenities $\rightarrow$ massive for low-income
- Small-to-big cities reallocation:
  - No wedges on earnings $\rightarrow$ largest effect for low-income Main result

Mechanism: Workers' Reallocation across Cities

	Baseline	Counterfactuals				
		Same Human Capital As Natives	Same Amenities As Natives	No Wedges On Earnings	Same Amenities As Natives & No Wedges On Earnings	Full
		(1)	(2)	(3)	(4)	(5)
Parameters						
$s_{okex} = s_{oUSex}$	-	x	-	-	-	x
$z_{jokex} = z_{joUSex}$	-	-	x	-	x	x
$\tau_{jok} = 1$	-	-	-	x	x	x
		Share Of Workers In The Big City				
Natives	82.0	-0.2	-0.4	-0.1	-0.5	-0.4
High-Income	82.8	-0.6	-1.5	0.5	-1.0	-1.1
Low-Income	90.0	-0.1	-12.3	1.2	-9.5	-9.6

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- Small-to-big cities reallocation:
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Mechanism: Workers' Reallocation across Cities

	Baseline	Counterfactuals				
		Same Human Capital As Natives	Same Amenities As Natives	No Wedges On Earnings	Same Amenities As Natives & No Wedges On Earnings	Full
		(1)	(2)	(3)	(4)	(5)
Parameters						
$s_{okex} = s_{oUSex}$	-	x	-	-	-	x
$z_{jokex} = z_{joUSex}$	-	-	x	-	x	x
$\tau_{jok} = 1$	-	-	-	x	x	x
		Share Of Workers In The Big City				
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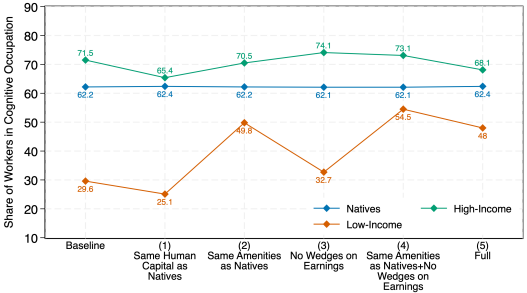
  - No differences in amenities → massive for low-income
- Small-to-big cities reallocation:
  - No wedges on earnings → largest effect for low-income Main result

Mechanism: Workers' Reallocation across Cities

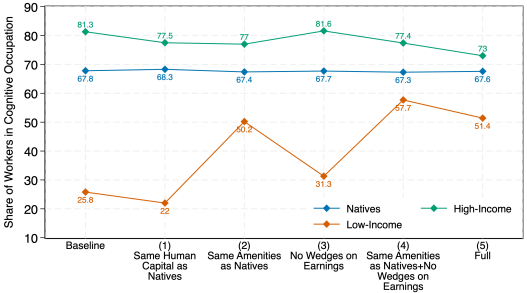
	Baseline	Counterfactuals				
		Same Human Capital As Natives	Same Amenities As Natives	No Wedges On Earnings	Same Amenities As Natives & No Wedges On Earnings	Full
		(1)	(2)	(3)	(4)	(5)
Parameters						
$s_{okex} = s_{oUSex}$	-	x	-	-	-	x
$z_{jokex} = z_{joUSex}$	-	-	x	-	x	x
$\tau_{jok} = 1$	-	-	-	x	x	x
		Share Of Workers In The Big City				
Natives	82.0	-0.2	-0.4	-0.1	-0.5	-0.4
High-Income	82.8	-0.6	-1.5	0.5	-1.0	-1.1
Low-Income	90.0	-0.1	-12.3	1.2	-9.5	-9.6

- Big-to-small cities reallocation:
  - No differences in human capital $\rightarrow$ workers from all countries
  - No differences in amenities $\rightarrow$ massive for low-income
- Small-to-big cities reallocation:
  - No wedges on earnings $\rightarrow$ largest effect for low-income Main result

Mechanism: Workers' Reallocation across Occupations



Small City



Big City

- No differences in human capital $\rightarrow$ immigrants in cognitive occupation $\downarrow$ in both cities
- No differences in amenities $\rightarrow$ low-income in cognitive occupation $\uparrow$ in both cities
- No wedges on earnings $\rightarrow$ immigrants in cognitive occupation $\uparrow$ in both cities

Main result

Mechanism: Competition Effect vs. Skills Effect

		Baseline	Counterfactuals				
			Same Human Capital As Natives	Same Amenities As Natives	No Wedges On Earnings	Same Amenities As Natives & No Wedge On Earnings	Full
			(1)	(2)	(3)	(4)	(5)
Parameters							
$s_{ojkex} = s_{oUSex}$		-	x	-	-	-	x
$z_{jokex} = z_{joUSex}$		-	-	x	-	x	x
$\tau_{jok} = 1$		-	-	-	x	x	x
				Small City			
Non-Cognitive	Competition	1	0.989	1.003	1.002	1.007	0.993
	Skills	1	1.040	0.983	1.005	0.993	1.041
Cognitive	Competition	1	1.004	0.999	0.999	0.998	1.002
	Skills	1	0.999	0.981	1.000	0.981	0.989
				Big City			
Non-Cognitive	Competition	1	0.978	1.018	1.004	1.023	1.008
	Skills	1	1.089	1.028	1.003	1.033	1.084
Cognitive	Competition	1	1.006	0.995	0.999	0.994	0.998
	Skills	1	1.001	0.990	0.998	0.986	0.992

- No differences in human capital $\rightarrow$ productivity $\uparrow$ in non-cognitive occupation in all cities
- No differences in amenities $\rightarrow$ productivity $\uparrow$ in non-cognitive occupations in the big city
- No wedges on earnings $\rightarrow$ no large changes in productivity in all cities Main result

Mechanism: Competition Effect vs. Skills Effect

		Baseline	Counterfactuals				
			Same Human Capital As Natives (1)	Same Amenities As Natives (2)	No Wedges On Earnings (3)	Same Amenities As Natives & No Wedge On Earnings (4)	Full (5)
Parameters							
$s_{ojkex} = s_{joUSex}$		-	x	-	-	-	x
$z_{jojkex} = z_{joUSex}$		-	-	x	-	x	x
$\tau_{jok} = 1$		-	-	-	x	x	x
				Small City			
Non-Cognitive	Competition	1	0.989	1.003	1.002	1.007	0.993
	Skills	1	1.040	0.983	1.005	0.993	1.041
Cognitive	Competition	1	1.004	0.999	0.999	0.998	1.002
	Skills	1	0.999	0.981	1.000	0.981	0.989
				Big City			
Non-Cognitive	Competition	1	0.978	1.018	1.004	1.023	1.008
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Cognitive	Competition	1	1.006	0.995	0.999	0.994	0.998
	Skills	1	1.001	0.990	0.998	0.986	0.992

- No differences in human capital $\rightarrow$ productivity $\uparrow$ in non-cognitive occupation in all cities
- No differences in amenities $\rightarrow$ productivity $\uparrow$ in non-cognitive occupations in the big city
- No wedges on earnings $\rightarrow$ no large changes in productivity in all cities Main result

Mechanism: Competition Effect vs. Skills Effect

		Baseline	Counterfactuals				
			Same Human Capital As Natives (1)	Same Amenities As Natives (2)	No Wedges On Earnings (3)	Same Amenities As Natives & No Wedge On Earnings (4)	Full (5)
Parameters							
$s_{ojkex} = s_{oUSex}$		-	x	-	-	-	x
$z_{jokex} = z_{joUSex}$		-	-	x	-	x	x
$\tau_{jok} = 1$		-	-	-	x	x	x
				Small City			
Non-Cognitive	Competition	1	0.989	1.003	1.002	1.007	0.993
	Skills	1	1.040	0.983	1.005	0.993	1.041
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- No differences in human capital $\rightarrow$ productivity $\uparrow$ in non-cognitive occupation in all cities
- No differences in amenities $\rightarrow$ productivity $\uparrow$ in non-cognitive occupations in the big city
- No wedges on earnings $\rightarrow$ no large changes in productivity in all cities Main result

Mechanism: Competition Effect vs. Skills Effect

		Baseline	Counterfactuals				
			Same Human Capital As Natives	Same Amenities As Natives	No Wedges On Earnings	Same Amenities As Natives & No Wedge On Earnings	Full
			(1)	(2)	(3)	(4)	(5)
Parameters							
$s_{ojkex} = s_{oUSex}$		-	x	-	-	-	x
$z_{jojkex} = z_{joUSex}$		-	-	x	-	x	x
$\tau_{jok} = 1$		-	-	-	x	x	x
				Small City			
Non-Cognitive	Competition	1	0.989	1.003	1.002	1.007	0.993
	Skills	1	1.040	0.983	1.005	0.993	1.041
Cognitive	Competition	1	1.004	0.999	0.999	0.998	1.002
	Skills	1	0.999	0.981	1.000	0.981	0.989
				Big City			
Non-Cognitive	Competition	1	0.978	1.018	1.004	1.023	1.008
	Skills	1	1.089	1.028	1.003	1.033	1.084
Cognitive	Competition	1	1.006	0.995	0.999	0.994	0.998
	Skills	1	1.001	0.990	0.998	0.986	0.992

- No differences in human capital $\rightarrow$ productivity $\uparrow$ in non-cognitive occupation in all cities
- No differences in amenities $\rightarrow$ productivity $\uparrow$ in non-cognitive occupations in the big city
- No wedges on earnings $\rightarrow$ no large changes in productivity in all cities Main result

Immigrants Amenities Estimates

Education	Small City		Big City	
	Non-Cognitive Occupation (1)	Cognitive Occupation (2)	Non-Cognitive Occupation (3)	Cognitive Occupation (4)
No College	1.0	0.4	7.3	2.1
	(0.0)	(0.3)	(4.4)	(0.8)
College	1.0	1.4	5.4	9.7
	(0.0)	(1.0)	(3.0)	(6.3)

Policy results

Immigrants Amenities Estimates

Education	Small City		Big City	
	Non-Cognitive Occupation (1)	Cognitive Occupation (2)	Non-Cognitive Occupation (3)	Cognitive Occupation (4)
No College	1.0	0.4	7.3	2.1
	(0.0)	(0.3)	(4.4)	(0.8)
College	1.0	1.4	5.4	9.7
	(0.0)	(1.0)	(3.0)	(6.3)

Policy results

Immigrants Amenities Estimates

Education	Small City		Big City	
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	(0.0)	(0.3)	(4.4)	(0.8)
College	1.0	1.4	5.4	9.7
	(0.0)	(1.0)	(3.0)	(6.3)

Policy results

Human Capital Estimates

Education	Occupation	Low-Income (1)	High-Income (2)	All Immigrants (3)
No College	Non-Cognitive	4.3	6.5	4.3
		(0.5)	(0.5)	(0.5)
	Cognitive	9.4	13.6	9.9
		(1.1)	(0.4)	(1.5)
College	Non-Cognitive	5.5	7.3	5.7
		(0.5)	(1.0)	(0.6)
	Cognitive	18.8	25.8	20.7
		(1.8)	(2.5)	(3.7)

Policy results

Human Capital Estimates

Education	Occupation	Low-Income (1)	High-Income (2)	All Immigrants (3)
No College	Non-Cognitive	4.3	6.5	4.3
		(0.5)	(0.5)	(0.5)
	Cognitive	9.4	13.6	9.9
		(1.1)	(0.4)	(1.5)
College	Non-Cognitive	5.5	7.3	5.7
		(0.5)	(1.0)	(0.6)
	Cognitive	18.8	25.8	20.7
		(1.8)	(2.5)	(3.7)

Policy results

Human Capital Estimates

Education	Occupation	Low-Income (1)	High-Income (2)	All Immigrants (3)
No College	Non-Cognitive	4.3	6.5	4.3
		(0.5)	(0.5)	(0.5)
	Cognitive	9.4	13.6	9.9
		(1.1)	(0.4)	(1.5)
College	Non-Cognitive	5.5	7.3	5.7
		(0.5)	(1.0)	(0.6)
	Cognitive	18.8	25.8	20.7
		(1.8)	(2.5)	(3.7)

Policy results

Immigration Policy Evaluation: Cities' Allocations

	Baseline	Policies	
		Inflow No College (1)	Inflow College (2)
		Employment Share	
Big City	82.8	+0.1	+0.1
		Cognitive Occupation Share	
Small City	3.8	+0.2	+0.8
Big City	5.4	+0.2	+1.1

Policy: Competition vs. Skills Effects

		Baseline	Policies	
			Inflow No College (1)	Inflow College (2)
			Small City	
Non-Cognitive	Competition	1	0.999	1.001
	Skills	1	0.996	0.999
Cognitive	Competition	1	1.000	1.000
	Skills	1	0.999	1.002
			Big City	
Non-Cognitive	Competition	1	0.997	1.001
	Skills	1	0.993	0.999
Cognitive	Competition	1	1.001	1.000
	Skills	1	0.999	1.003

i. Competition and skills effect larger in big cities than in small cities
ii. In all cities, positive skills effect, while competition effect is negligible

Policy

Policy: Competition vs. Skills Effects

		Baseline	Policies	
			Inflow No College (1)	Inflow College (2)
			Small City	
Non-Cognitive	Competition	1	0.999	1.001
	Skills	1	0.996	0.999
Cognitive	Competition	1	1.000	1.000
	Skills	1	0.999	1.002
			Big City	
Non-Cognitive	Competition	1	0.997	1.001
	Skills	1	0.993	0.999
Cognitive	Competition	1	1.001	1.000
	Skills	1	0.999	1.003

i. Competition and skills effect larger in big cities than in small cities
ii. In all cities, positive skills effect, while competition effect is negligible Policy

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